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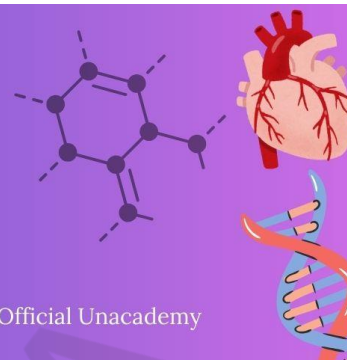
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LAST 10 YEARS

NEET PYQ



Chapter 11 : Ecosystem

1. Given below are two statements:

[NEET-2025]

Statement I : The primary source of energy in an ecosystem is solar energy.

Statement II : The rate of production of organic matter during photosynthesis in an ecosystem is called net primary productivity (NPP).

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

2. Who is known as the father of Ecology in India?

[NEET-2025]

- (1) S.R. Kashyap
- (2) Ramdeo Misra
- (3) Ram Udar
- (4) Birbal Sahni

3. Given below are two statements:

[NEET-2025]

Statement I: In ecosystem, there is unidirectional flow of energy from the sun from producers to consumers.

Statement II: Ecosystems are exempted from 2nd law of thermodynamics.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

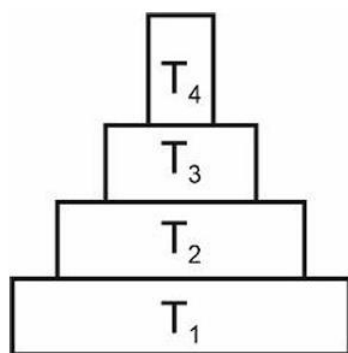
4. Which of the following is the unit of productivity of an Ecosystem?

[NEET-2025]

- (1) gm^{-2}
- (2) KCal m^{-2}
- (3) KCal m^{-3}
- (4) $(\text{KCal m}^{-2})\text{yr}^{-1}$

5. Consider the pyramid of energy of an ecosystem given below:

[Re-NEET-2024]



If T_4 is equivalent to 1000 J, what is the value at T_1

- (1) 10000 / 10 J (2) 10000 / 10 × 4 J
(3) 10,000 J (4) 10,00,000 J

6. Which one of the following is not a limitation of ecological pyramids?

[Re-NEET-2024]

- (1) Saprophytes are not given any place of ecological pyramids
- (2) It assumes a simple food chain, that almost never exists in nature
- (3) It accommodates a food web
- (4) It does not take into account the same species belonging to two or more trophic levels

7. In an ecosystem if the Net Primary Productivity (NPP) of first trophic level is $100x \text{ (kcal m}^{-2} \text{ yr}^{-1}\text{)}$, what would be the GPP (Gross Primary Productivity) of third trophic level of the same ecosystem?

- $$(1) \quad \frac{x}{10} (\text{kcal m}^{-2}) \text{yr}^{-1}$$

[NEET-2024]

- (2) $x(\text{kcal m}^{-2})\text{yr}^{-1}$

- (3) $10x(\text{kcal m}^{-2})\text{yr}^{-1}$

- (4) $\frac{100x}{3x} (\text{kcal m}^{-2}) \text{yr}^{-1}$

8. The amount of nutrients such as carbon, nitrogen, potassium and calcium present in the soil at any given time is referred to as : **[NEET-2023-MANIPUR]**

[NEET-2023-MANIPUR]

- (1) Humus (2) Detritus
(3) Standing state (4) Standing crop

9. In the equation $GPP - R = NPP$

[NEET-2023]

GPP is Gross Primary Productivity ; NPP is Net Primary Productivity

R here is _____.

- (1) Photosynthetically active radiation
- (2) Respiratory quotient
- (3) Respiratory loss
- (4) Reproductive allocation

10. Identify the correct statements:

[NEET-2023]

A. Detritivores perform fragmentation.

B. The humus is further degraded by some microbes during mineralization.

C. Water soluble inorganic nutrients go down into the soil and get precipitated by a process called leaching. D. The detritus food chain begins with living organisms.

E. Earthworms break down detritus into smaller particles by a process called catabolism.

Choose the correct answer from the options given below:

(1) A, B, C only

(2) B, C, D only

(3) C, D, E only

(4) D, E, A only

11. The amount of biomass or organic matter produced per unit area over a time period by plants during photosynthesis is called:

[NEET-2022]

(1) Secondary production

(2) Primary production

(3) Gross primary production

(4) Net primary production

12. Match the List-I with List-II

[NEET-2022]

List-I

List-II

(a) Carbon dissolved in oceans

(i) 55 billion tons

(b) Annual fixation of carbon through photosynthesis

(ii) 71%

(c) PAR captured by plants

(iii) 4×10^{13} kg

(d) Productivity of oceans

(iv) 2 to 10%

Choose the correct answer from the options given below:

(1) (a) - (ii), (b) - (iv), (c) - (iii), (d) - (i)

(2) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

(3) (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)

(4) (a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)

13. The amount of nutrients, such as carbon, nitrogen, phosphorus and calcium present in the soil at any given time, is referred as :-

[NEET-2021]

(1) Climax

(2) Climax community

(3) Standing state

(4) Standing crop

14. Which of the following statements is not correct?

[NEET-2021]

(1) Pyramid of biomass in sea is generally inverted.

(2) Pyramid of biomass in sea is generally upright.

(3) Pyramid of energy is always upright.

(4) Pyramid of numbers in a grassland ecosystem is upright.

15. In the equation $GPP - R = NPP$ R represents :- **[NEET-2021]**
- (1) Radiant energy
 - (2) Retardation factor
 - (3) Environment factor
 - (4) Respiration losses
16. In relation of Gross primary productivity and Net primary productivity of an ecosystem, which one of the following statements is correct? **[NEET-2020]**
- (1) There is no relationship between Gross primary productivity and Net primary productivity.
 - (2) Gross primary productivity is always less than net primary productivity.
 - (3) Gross primary productivity is always more than net primary productivity.
 - (4) Gross primary productivity and Net primary productivity are one and same.
17. Which of the following ecological pyramids is generally inverted? **[NEET-2019]**
- (1) Pyramid of numbers in grassland
 - (2) Pyramid of energy
 - (3) Pyramid of biomass in a forest
 - (4) Pyramid of biomass in a sea
18. Which of the following statements is incorrect? **[NEET-2019]**
- (1) Biomass decreases from first to fourth trophic level
 - (2) Energy content gradually increases from first to fourth trophic level
 - (3) Number of individuals decreases from first trophic level to fourth trophic level
 - (4) Energy content gradually decreases from first to fourth trophic level
19. The rate decomposition is faster in the ecosystem due to following factors EXCEPT **[NEET-2019]**
- (1) Detritus rich in sugars
 - (2) Warm and moist environment
 - (3) Presence of aerobic soil microbes
 - (4) Detritus richer in lignin and chitin
20. Which ecosystem has the maximum biomass? **[NEET-2017]**
- (1) Grassland ecosystem
 - (2) Pond ecosystem
 - (3) Lake ecosystem
 - (4) Forest ecosystem
21. The sequential events from initial stage till climax stage in a succession are called :- **[NEET-2017]**
- (1) Ecesis
 - (2) Sere
 - (3) Nudation
 - (4) Migration

22. The term ecosystem was coined by :-

[NEET- I 2016]

- (1) E. P. Odum
- (2) A. G. Tansley
- (3) E. Haeckel
- (4) E. Warming

23. The primary producers of the deep-sea hydrothermal vent ecosystem are :-

[NEET- II 2016]

- (1) Blue-green algae
- (2) Coral reefs
- (3) Green algae
- (4) Chemosynthetic bacteria

ANSWER KEY

1. (3)
2. (2)
3. (3)
4. (4)
5. (4)
6. (3)
7. (3)
8. (3)
9. (3)
10. (1)
11. (2)
12. (3)
13. (3)
14. (2)
15. (4)
16. (3)
17. (4)
18. (2)
19. (4)
20. (2)
21. (4)
22. (2)
23. (2)